FAJAR NUGRAHA, M.Si.

Soil Scientist | GIS & Machine Learning Specialist
Makassar, South Sulawesi

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Master's graduate in Soil Science from IPB with a specialization in spatial analysis, watershed hydrology, and machine learning for land management. Experienced in research, geospatial data processing, and WebGIS development based on environmental analysis. Accustomed to integrating scientific and technological approaches (GIS, Python, GEE) to support data-driven decision-making in land and environmental resource management.

TECHNICAL SKILLS

- **Core Soil Science:** Soil & Water Conservation, Digital Soil Mapping, Soil Physics, Land Suitability Evaluation, Watershed Management.
- **GIS & Remote Sensing:** ArcGIS, QGIS, Google Earth Engine (GEE), GlobalMapper, Spatial-Temporal Analysis.
- **Data Science & Programming:** Python (Scikit-learn, Pandas), R, Machine Learning, JavaScript (NodeJS), HTML/CSS.

EXPERIENCE

Support Staff - Meta Information Technology Juni – Desember 2026
East Jakarta, DKI Jakarta

- Prepared and analyzed regional vulnerability data by integrating Social, Physical, Economic, and Environmental parameters for the National Agency for Disaster Countermeasure (BNPB).
- Prepared and analyzed climate change vulnerability data for SIDIK (Vulnerability Index Data Information System) for the Ministry of Environment (KLH).
- Responsible for the preparation of outputs and final report documents for vulnerability mapping and SIDIK.

Practicum Assistant Coordinator, Department of Soil Science - Hasanuddin University 2021 - 2022, 2024
Makassar, South Sulawesi

- Coordinated all practicum assistants for the Fundamentals of Soil Science, Soil and Water Conservation, and Soil and Water Management courses.
- Ensured the smooth execution of practicums for approximately 900 students from 3 different study programs.
- Acted as the main liaison between practicum assistants and supervising lecturers to communicate practicum needs and logistics.
- Provided technical briefings regarding practicum procedures to the assistant team and students.

Practicum Assistant, Department of Soil Science - Hasanuddin University 2021 - 2022
Makassar, South Sulawesi

- Assisted and provided technical guidance for the Land Resource Information Systems, Soil Physics, Remote Sensing, and Agricultural Hydrology courses.
- Supervised and provided objective assessments to students during practicum activities.
- Reported any obstacles and practicum progress periodically to the Assistant Coordinator.

PROJECTS

Erosion Analysis WebGIS (USLE) - Fullstack Mar 2026

- **Technical Focus:** Automation of complex spatial mathematical calculations.
- Designed a fullstack WebGIS architecture to automatically calculate the rate of surface erosion using the empirical USLE (Universal Soil Loss Equation).
- Integrated spatial data processing on the backend using NodeJS.
- Dynamically calculated the overlay of various biophysical parameters, including rainfall factor, soil erodibility, slope length, land cover, and conservation practices.
- Presented real-time computation results in the form of interactive erosion hazard mapping, significantly cutting manual analysis time.

Spectral Index Analysis WebGIS Feb 2026

- **Technical Focus:** Satellite image processing and environmental data science.
- Built a web-based spatial intelligence application to instantly extract and evaluate vegetation health through multispectral image processing.
- Utilized the Python programming language as a computational engine to execute the NDVI algorithm and other spatial environmental index variants.
- Developed an interactive interface that visualizes the analysis results of land degradation detection and forest canopy density with high accuracy.

Interactive Dashboard WebGIS & Environmental Infographics Jan 2026

- **Technical Focus:** Big Data Visualization and User Experience (UX) Data.
- Developed a frontend spatial analytic dashboard specifically designed to visualize geospatial and ecological big data.
- Translated complex environmental technical data into interactive infographics that are easily understood by policymakers (stakeholders).
- Optimized browser rendering performance for dynamic and responsive thematic mapping and interactive spatial data filters.

Soil Science Calculator Website Jan 2026

- **Technical Focus:** Digitalization of academic formulas into functional products.
- Translated complex mathematical formulas of soil physics and chemistry into a JavaScript-based computational algorithm.
- Created a mobile-friendly digital calculator tool that serves as a field aid for researchers and agronomists.
- Facilitated instant calculations for crucial soil parameters, such as porosity values, bulk density, water content, and ameliorant requirement estimations.

Web Developer - Lontara.tech Dec 2026

- **Technical Focus:** End-to-End software project management.
- Orchestrated the entire Software Development Life Cycle for a digital portfolio platform and spatial project repository.
- Wrote pure HTML5, CSS3, and JavaScript source code from scratch to build an adaptive and responsive cross-device interface.
- Managed the deployment process and system integration independently, optimizing accessibility, load speed, and central hub function for independent WebGIS experiments.

EDUCATION

Master of Science (M.Si), Soil Science – IPB University (GPA: 3.83/4.00) | 2023-2025
Bogor, West Java

- **Thesis:** Land Use Directions to Improve Watershed Performance Using Machine Learning in the Tanralili Sub-watershed.

Bachelor of Agriculture (S.P), Agrotechnology – Hasanuddin University (GPA: 3.90) | 2018-2022
Makassar, South Sulawesi

- **Thesis:** The Effect of Land Cover Changes on River Flow Components in the Tanralili Sub-watershed.
- **Awards:** Best Graduate of the Soil Science Department (December 2022).

SCIENTIFIC PUBLICATIONS

- Evaluation of Machine Learning Algorithms for Land Use Classification and Prediction (Sinta 2). 2025. https://doi.org/10.22146/mgi.99150.
- Hydrograph Modeling to Assess the Impact of Land Use Changes in the Tanralili Sub-Watershed (Sinta 2). 2025. https://doi.org/10.59465/jppdas.2025.9.1.1-22.

SERTIFIKASI

- Fundamental Python | Coding Studio | July 2024
- Fundamental Data Science | Coding Studio | April 2024
- Data Science and Machine Learning with R | Udemy | April 2024

PRESTASI

- 3rd Runner Up, National Scientific Writing Competition, Brawijaya Machine Creativity (Internet of Things (IoT) Category), October 2022.
- Bronze Medal, Scientific Writing Competition, 6th International Avicenna Youth Science Fair (Chemistry Category), October 2021.
- Zone Finalist, Independent Young Entrepreneur Business Competition (Technology Business Plan Category), November 2020.